

Segmenting ads24x7's Digital Ad Campaigns

A three-segment K-Means model of campaign efficiency, built to guide how ads24x7 deploys its new \$10M seed budget

Executive Summary

This report segments 25,857 digital ad campaigns run by ads24x7, which recently raised \$10 million in seed funding and wants an evidence-based view of where that money should go. We reconstructed the three efficiency metrics that were missing from parts of the dataset (CTR, CPM, CPC) from their underlying formulas, capped a small number of extreme values so they could not distort the clustering, standardized every feature, and compared Ward's hierarchical method against K-Means before settling on a **three-segment K-Means model**.

Two findings drive the recommendation. First, the segment consuming the largest share of budget is also the least efficient: it takes **61% of all spend** to deliver a 0.24% click-through rate and clicks costing ₹0.78 — eight times what the efficient segment pays. Second, the thing that separates the segments is not device or platform but **inventory type**, and it does so almost deterministically: not one of the 10,763 Format1, Format2 or Format3 ads lands in the efficient segment, while 98% of Format4 ads do. That gives ads24x7 a concrete lever rather than a general observation.

Acting on that is testable rather than speculative. Reallocating the seed budget toward the efficient segment projects **55% more clicks than the current spend pattern buys, while deploying 10% less of the money** — Section 8 sets out that arithmetic and the conditions attached to it.

1. Marketing Metric Definitions

Three formulas do the heavy lifting, each derived from Spend, Impressions and Clicks. These three are also the features we cluster on: they are the numbers the media-buying team already trades on, and unlike raw Spend or Clicks they describe how *efficiently* an ad runs rather than how big it is.

Metric	Formula	What it tells us
CPM (Cost per Mille)	$(\text{Spend} / \text{Impressions}) \times 1,000$	What it costs to put an ad in front of 1,000 people — how expensive the inventory is to buy in bulk.
CPC (Cost per Click)	$\text{Spend} / \text{Clicks}$	What ads24x7 pays each time someone actually clicks — the number that matters once engagement, not reach, is the goal.
CTR (Click-Through Rate)	$(\text{Clicks} / \text{Impressions}) \times 100$	The share of views that become a click — a proxy for whether the creative and targeting are landing.

2. Data Overview and Missing Value Imputation

The dataset holds 25,857 campaign-level rows across 19 columns, with no duplicate rows. Three derived metrics were incomplete: CTR missing in 6,465 rows, CPM in 6,465, and CPC in 7,527. Rather than fill these with a mean or median — which would invent performance that never happened — or drop the rows, a single function recomputes each missing value from Spend, Impressions and Clicks using the formulas above. Every one of the 25,857 campaigns was retained.

Two properties of the raw data that had to be handled

The supplied CTR values are not on the scale the data dictionary specifies. The dictionary defines CTR as (Clicks / Impressions) × 100, but the values already in the file are the raw fraction Clicks / Impressions. We tested this rather than assuming it: the median ratio of each stored value to Clicks / Impressions is 1.0000, not 100. Imputing the missing cells with the documented formula while leaving the supplied ones untouched would give a column that is 25% percentages and 75% fractions — values a hundredfold apart inside a single feature. Because clustering is a distance calculation, that is not a cosmetic problem; it would make the CTR axis meaningless and the segments arbitrary. We therefore rescale the whole column onto the documented percentage basis before anything else happens to it, and every CTR figure in this report is on that basis.

Zero denominators needed an explicit rule. CPM and CTR divide by Impressions and CPC divides by Clicks, so some rows are mathematically undefined rather than merely missing: 219 rows have zero impressions and 2,791 have zero clicks. We checked what those rows actually are before deciding. All 219 zero-impression rows also have zero spend and zero clicks — dormant ads that never served. Of the 2,791 zero-click rows, 2,586 also have zero spend, and the remaining 205 spent at most ₹0.19 in total. Setting these to 0 is therefore factually right rather than a convenient filler, and it avoids infinities entering a distance calculation. One further row record 2 clicks against 1 impression, which is impossible; it is a tracking artefact and is capped in Section 3.

Metric	Mean	Std Dev	Median	IQR (Q1–Q3)	Max
Clicks	9,525.88	16,721.69	3,457.00	305 – 10,681	143,049
Spend (₹)	2,414.47	3,932.84	1,173.66	36.03 – 2,692.28	26,931.87
CPM (₹)	7.52	8.93	2.96	1.57 – 12.45	715.00
CTR (%)	7.50	9.13	0.72	0.23 – 13.06	200.00
CPC (₹)	0.30	0.34	0.12	0.08 – 0.50	7.26

Table 1: Summary statistics after imputation, before outlier treatment. CTR is on the documented percentage basis described above.

3. Handling Outliers Before Clustering

K-Means represents each cluster by the mean position of its members and minimizes squared distance to it. Squaring means a point ten times further out contributes a hundred times more error, and a mean has no breakdown point, so a handful of extreme campaigns can drag a centroid away from the bulk of its members — or capture a cluster outright, producing one oversized group plus a tiny cluster of anomalies that tells nobody anything.

Deleting those campaigns is the wrong fix. Ad performance is genuinely right-skewed: premium placements really do cost twenty times a remnant impression, and the extreme rows are real business activity, not data errors. Removing them would bias the whole analysis toward cheap inventory. We therefore **cap rather than delete**, wins rising each feature at its 1st and 99th percentiles. Every row is kept and rank order is preserved — the most expensive ad stays the most expensive ad, it simply stops exerting leverage proportional to how far out it sits — and the features stay in their original units, so the profile table in Section 6 reads directly as money.

Feature	IQR outliers flagged	Cap applied (1st / 99th pct)	Values capped	Skew before	Skew after
CPM	229 (0.89%)	0.00 / 27.66	259 (1.00%)	21.00	0.74
CPC	710 (2.75%)	0.00 / 1.48	256 (0.99%)	1.89	1.42
CTR	302 (1.17%)	0.00 / 33.33	220 (0.85%)	2.49	0.88

Table 2: Outliers identified by the 1.5×IQR rule, and the effect of capping. Roughly 1% of values move in each feature; CPM's skew falls from 21.0 to 0.7.

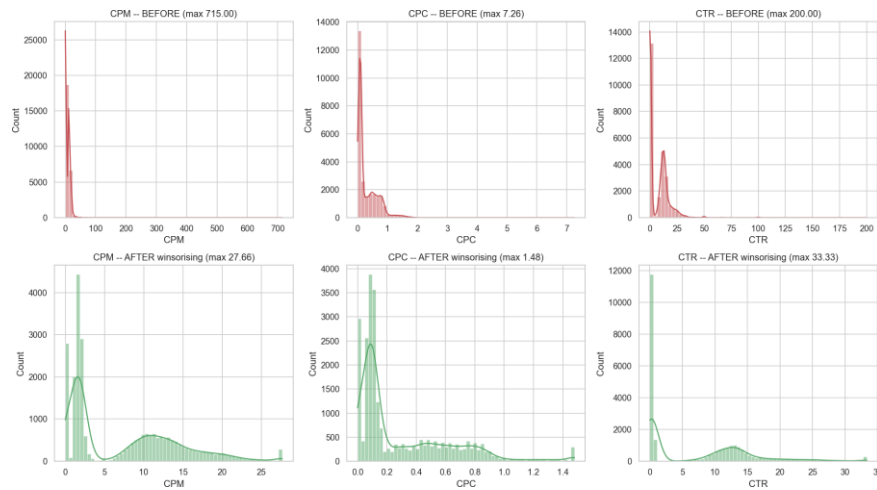


Figure 1: Distribution of each feature before (top) and after (bottom) winsorisation. The shape of the data is preserved; only the extreme tail is pulled in.

4. Why We Standardized the Features

After capping, each feature was rescaled to a mean of 0 and a standard deviation of 1. Two reasons. First, without it one feature drowns out the rest: CPM runs to roughly ₹28 while CTR sits mostly below 1%, so a ₹10 CPM gap would contribute about a hundred times more to the squared distance than the entire range of CTR, and the clusters would be CPM bands with the other two features along for the ride. Standardizing puts all three into the same unit — standard deviations from their own average — so each contributes comparably.

Second, it helps convergence. When features sit on wildly different scales the error surface is stretched along one axis, and the algorithm spends iterations inching along that dominant direction. On comparably scaled axes the surface is closer to spherical and the search is better behaved.

5. Choosing a Clustering Method and a Value of k

Hierarchical clustering runs on the full dataset. Ward's method is often sampled down on the assumption that it needs a full pairwise distance matrix, which for 25,857 records would be roughly 2.6GB. That is true only of a precomputed distance matrix. `scipy.linkage(X, method='ward')` instead uses the nearest-neighbour-chain algorithm, which is $O(n^2)$ in time but only $O(n)$ in memory, and it completes on all 25,857 records in under a minute. We therefore build the dendrogram on the entire dataset, so the structure it shows is the structure K-Means is being asked to find rather than an artefact of a sample.

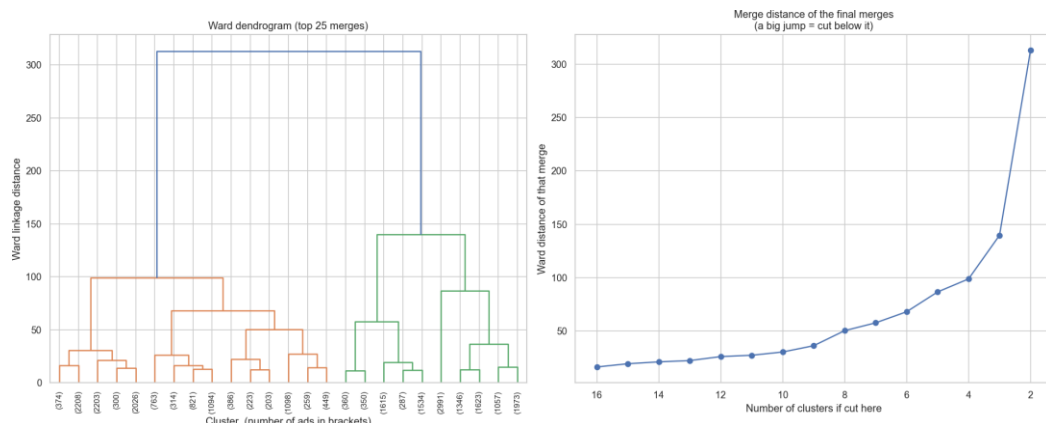


Figure 2: Ward dendrogram on all 25,857 records (left, top 25 merges) and the merge distance of the final merges (right). The tree splits into two branches at a distance of ~313, and those split again at ~140 and ~99.

The dendrogram supports a cut at 2, 3 or 4 clusters: a horizontal line between 100 and 139 crosses exactly three vertical lines. The elbow plot agrees on the region — WCSS falls steeply and flattens around 3 to 4 — while the average silhouette is highest at $k=2$. Three diagnostics, three answers.

k	Inertia (WCSS)	Average silhouette	Weakest cluster's silhouette	Worst cluster: % of members on a boundary
2	28,610	0.599	0.580	0.8%
3 (our choice)	17,907	0.555	0.549	14.3%
4	12,464	0.513	0.206	53.0%
5	9,424	0.520	0.204	53.6%

Table 3: k compared on the average silhouette and on the health of its worst cluster. 'On a boundary' means a member scoring below 0.25, i.e. nearly equidistant from its own cluster and the next one.

The diagnostic that settles it

An average silhouette is a single number for the whole partition, and it hides the failure that matters most: a solution can post a respectable average while containing one cluster that is not really a cluster — a slice carved off a neighbor, whose members sit on the boundary between the two. Three healthy clusters will happily average over one bad one. So, we looked inside each candidate, at every cluster's own silhouette and at the share of its members scoring below 0.25.

That is what rules out k=4. Its average of 0.513 looks acceptable, but the fourth cluster scores 0.206 and **53% of its members fall below 0.25** — more than half of it sits on the boundary with the cluster it was split from. It is a gradient within the efficient segment, not a mode of its own. k=5 repeats the same failure.

k=2 is rejected for a different and more specific reason. It is the cleanest split on every purely statistical measure, but it merges the two low-CTR segments into one group holding 71% of spend. That collapse hides the central finding of the analysis — that one segment burns 61% of the budget at a 0.24% CTR while a structurally different segment spends 10% at a quarter the cost per click. At k=2 those two are the same cluster and the recommendation degenerates into 'some ads are efficient and some are not'.

At k=3 every cluster scores between 0.549 and 0.570 with 3–14% of members near a boundary: no weak cluster anywhere in the solution. It also minimizes the Davies–Bouldin index. Moving from k=2 to k=3 costs little (0.599 to 0.555) and buys the separation the business decision depends on. On this feature set the statistics support k=3 directly; it does not have to be argued for against the numbers.

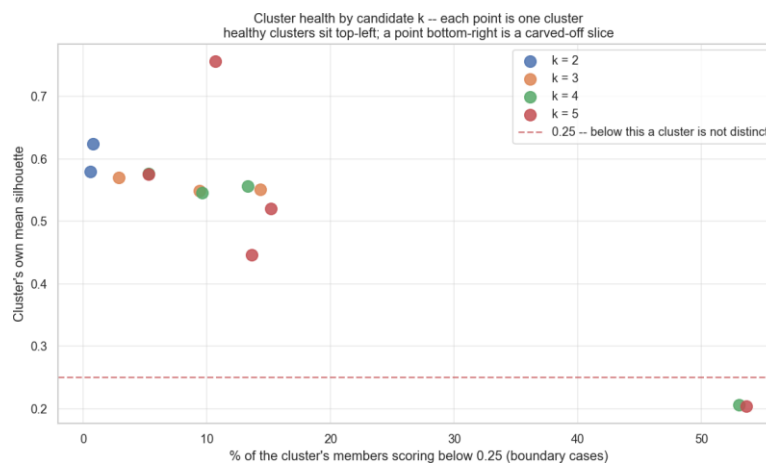


Figure 3: Every cluster in every candidate solution, plotted by its own silhouette against the share of its members sitting on a boundary. Healthy clusters sit top-left. The two points in the bottom-right are the fourth and fifth clusters — the ones that are not really clusters.

6. What Each Segment Looks Like

Splitting the portfolio into three produces segments that differ sharply on cost and on conversion:

Metric performer	Efficient	High-reach, low-engagement	Cheap-impression volume
Campaign count (%) (48.6%)	12,572	6,661 (25.8%)	6,624 (25.6%)
Share of total spend	28.8%	61.3%	10.0%
Mean CPM (₹)	13.	1.75	1.16
Mean CPC (₹)	0.	0.78	0.19
Mean CTR (%)	14.	0.24	0.42
Mean spend (₹)	1,428	5,744	939
Mean impressions	108,358	3,473,482	624,721
Mean clicks	14,330	7,250	2,696

Table 4: Segment profiles in original, unscaled units.

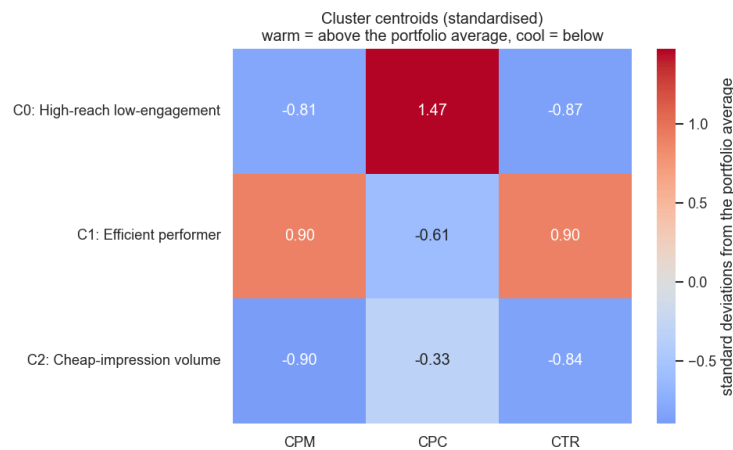


Figure 4: Segment centroids in standard deviations from the portfolio average. Warm is above average, cool is below — a neutral scale, since a high CPC is expensive rather than good.

Efficient performer — 48.6% of campaigns, 28.8% of spend

The commercial core of the business. A 14.74% CTR at ₹0.10 per click, on mid-priced reach. It generates more clicks per campaign (14,330) than either other segment while spending a quarter of what the largest segment spends. This is the scaling candidate.

High-reach, low-engagement — 25.8% of campaigns, 61.3% of spend

Enormous delivery, averaging 3.5 million impressions per campaign at a cheap ₹1.75 CPM, but a 0.24% CTR and the most expensive clicks in the portfolio at ₹0.78 — eight times the efficient segment. It consumes ₹38.3M, nearly two-thirds of all spend, and still returns fewer clicks per campaign than the efficient segment does. This is the single largest margin opportunity in the analysis.

Cheap-impression volume — 25.6% of campaigns, 10.0% of spend

The cheapest impressions available at ₹1.16 CPM and real volume, 624,721 impressions per campaign. Conversion is weak at 0.42%, but clicks still cost only ₹0.19 — a quarter of what the high-reach segment pays. The contrast between these two is instructive: same weak engagement, four times the click efficiency.

The premium tail, reported as a flag rather than a fourth segment

Rejecting $k=4$ does not mean the high-performing tail it tried to isolate is imaginary, so we flag it instead. One caution had to be handled first: CTR is a ratio, and ratios on tiny denominators are unstable — an ad with 4 impressions and 1 click posts a 25% CTR that means nothing. Below about 1,000 impressions mean and median CTR diverge sharply (roughly 10% against 0%), which is the signature of exactly that noise. Applying a

10,000-impression floor and taking the top decile of the efficient segment by CTR gives **671 campaigns averaging an 18.64% CTR at ₹0.085 per click on a median 68,043 impressions** — real delivery, sustained performance, 1.8% of spend. Without the floor the same rule returns mostly sub-1,000-impression ads worth 0.09% of spend, which would have been misleading.

What Separates the Segments: Inventory Type, Not Device

Device type explains none of the difference. Ads run 64% on mobile and 36% on desktop, and that split is near-identical inside every segment (35.6–35.9% desktop across all three). Platform, format and ad type are equally flat. Whatever drives the gap between segments affects both device types equally, so device is not a useful lens for targeting decisions.

Inventory type is a completely different story, and it is the most actionable finding in this analysis:

Inventory type	Ads	→ Efficient performer	→ High-reach, low-engagement	→ Cheap-impression volume
Format1	5,432	0.0%	60.4%	39.6%
Format2	1,789	0.0%	100.0%	0.0%
Format3	3,542	0.0%	23.3%	76.7%
Format4	7,218	98.2%	0.0%	1.7%
Format5	5,151	81.4%	0.1%	18.5%
Format6	1,968	48.1%	28.2%	23.7%
Format7	757	45.0%	27.1%	27.9%

Table 5: Where each inventory type lands. Rows sum to 100%.

The mapping is close to deterministic. **Not one of the 10,763 Format1, Format2 or Format3 campaigns lands in the efficient segment** — not a single one — while 98% of Format4 and 81% of Format5 campaigns do. Format2 sits entirely in the worst-performing segment. Efficiency in this portfolio is, to a first approximation, a property of which inventory was bought.

Format1 is worth a note, because it is the largest single inventory type and it appears in both weak segments. That is not inconsistency: it never reaches the efficient segment at all. It is consistently poor, and the 60/40 split between the two low-CTR segments reflects the price paid for it rather than any difference in how it performs.

7. Recommended Use of the \$10M Seed Budget

The segment profile in Section 6 reports averages per campaign. A budget decision needs a different quantity: what each segment returns per unit of money put into it. That is a ratio of totals, not an average of per-campaign ratios — averaging campaign-level CPCs would weight a ₹1 campaign the same as a ₹20,000 one. Computed on totals, the three segments are not close:

Segment	Share of spend	Share of clicks	Clicks per ₹1,000 spent	Blended CPC (₹)
Efficient performer	28.7%	73.1%	10,038	0.10
High-reach, low-engagement	61.3%	19.6%	1,262	0.79
Cheap-impression volume	10.0%	7.2%	2,870	0.35

Table 6: Segment efficiency computed on totals across the full dataset.

The efficient segment takes 28.7% of spend and returns 73.1% of clicks. The high-reach segment does the reverse: 61.3% of spend for 19.6% of clicks. Per rupee, it buys **8.0 times fewer clicks** — 1,262 against 10,038 per ₹1,000. The cheap-impression segment sits between them at 2,870.

Why revenue share cannot justify the allocation

It is tempting to defend the high-reach segment on the grounds that it generates 62.8% of recorded revenue. That figure is real but circular. Revenue in this dataset is Spend net of a Fee that ranges only from 0.21 to 0.35, so revenue correlates with spend at 0.9987 and returns ₹0.66–₹0.73 per rupee spent in every segment. A

segment's revenue share is therefore a restatement of its spend share and carries no information about whether that spending worked. Clicks per rupee does, which is why the allocation below is built on it.

The allocation

Segment	Allocation	Basis
Efficient performer	\$5.5M (55%)	Returns 10,038 clicks per ₹1,000 — the best in the portfolio by a factor of eight. \$5.5M on top of its ₹17.9M historical spend is a 31% scale-up rather than a leap into untested territory, and it is concentrated in Format4 and Format5 inventory, where 98% and 81% of that inventory already lands in this segment.
High-reach, low-engagement	\$2.5M (25%), conditional	Cut from its current 61.3% share. At 1,262 clicks per ₹1,000 it cannot be justified on performance, but it may be running to a reach objective rather than a click one. Release this tranche only once that is established; if it is a performance buy, it is failing and the money belongs in the efficient segment.
Cheap-impression volume	\$1.0M (10%)	At ₹1.16 CPM this is the cheapest place in the portfolio to run creative and targeting tests. It also supplies the direct comparison that prices the high-reach segment: same weak CTR, 2.3 times the clicks per rupee.
Held back	\$1.0M (10%)	Not allocated to a segment. Funds the two open questions this analysis raises — whether the 671 flagged high-CTR campaigns can be sourced at greater volume, and what the high-reach segment is actually for — and periodic re-running of the segmentation as the ad mix shifts.

Table 7: Allocation of the \$10M seed budget.

Holding each segment's current efficiency constant, this split projects **61.2 million clicks against 39.4 million** for the same \$10M spread in today's proportions — roughly 55% more, while leaving 10% of the budget undeployed. That projection assumes efficiency does not decay as the efficient segment grows, which is precisely why the recommendation pairs the shift with a delivery test rather than treating the number as a promise.

8. Conclusion

Three segments describe this portfolio, and they are separated by efficiency rather than by size. Just under half the campaigns sit in an efficient segment converting at 14.74% for ₹0.10 a click; a quarter sit in a high-reach segment converting at 0.24% for ₹0.79; the remaining quarter buy the cheapest impressions available and convert weakly but cheaply. Reconstructing the missing metrics from their formulas kept all 25,857 campaigns in the analysis, and capping rather than deleting outliers kept the genuine high-value activity that a deletion rule would have discarded.

On the number of segments, the average silhouette and the elbow point in different directions, and the disagreement is settled by asking whether every cluster in a solution is actually a cluster. At $k=4$ one is not: it scores 0.206 with over half its members on a boundary. At $k=3$ all three score between 0.549 and 0.570. Three segments won on that evidence rather than on being the more convenient number to work with.

Two findings carry the recommendation. The first is that the largest consumer of budget is the least efficient: 61.3% of spend returns 19.6% of clicks, eight times worse per rupee than the efficient segment, and the revenue figures that might appear to defend it are a mechanical restatement of its own spend. The second is that inventory type predicts which segment a campaign land in almost perfectly — no Format1, Format2 or Format3 campaign reaches the efficient segment, while 98% of Format4 does. Together they make the first move concrete: establish what the high-reach segment is bought to achieve, and shift budget toward the inventory that has already demonstrated it converts.

